

CYBERNAUT ED TECH PROJECT REPORT

WEATHER FORECAST APPLICATION USING DJANGO

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PROJECT TITLE

WEATHER FORECAST APPLICATION USING DJANGO

PROJECT DOMAIN

Web Application Development

Technology Stack

Python | Django | HTML | CSS | JavaScript

Submitted To

CYBERNAUT EDTECH

INTRODUCTION

The Weather Forecast Web Application is a web-based application developed using Python and Django Framework to provide users with real-time weather information and city-based weather forecasting.

Weather plays an important role in daily life, travel, agriculture, transportation, and planning activities.

Traditional methods of obtaining weather information require users to switch between multiple websites and applications. This project aims to provide a centralized platform where users can search weather conditions instantly and maintain their weather search history.

The application provides an interactive user interface with login functionality and weather visualization. Users can create accounts, log in securely, search weather conditions for different cities, and view previous search history.

The weather information is collected dynamically using the OpenWeather API and displayed in a visually attractive dashboard.

Main modules included in the system:

- Home Module
- User Module
- Weather Search Module
- History Module
- Authentication Module

OBJECTIVES

The objectives of the Weather Forecast Application are:

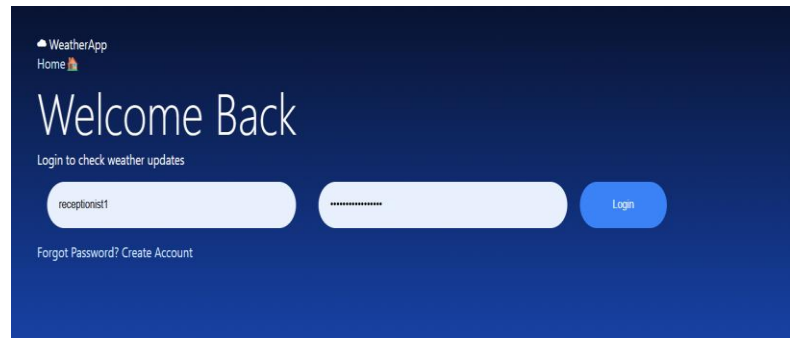
- To provide real-time weather information.
- To allow users to search weather conditions for any city.
- To display temperature and weather status clearly.
- To maintain search history.
- To implement user registration and login.
- To improve user experience using responsive UI.
- To reduce dependency on external weather websites.
- To create a scalable weather platform using Django.

SYSTEM MODULES

A. HOME MODULE

Features:

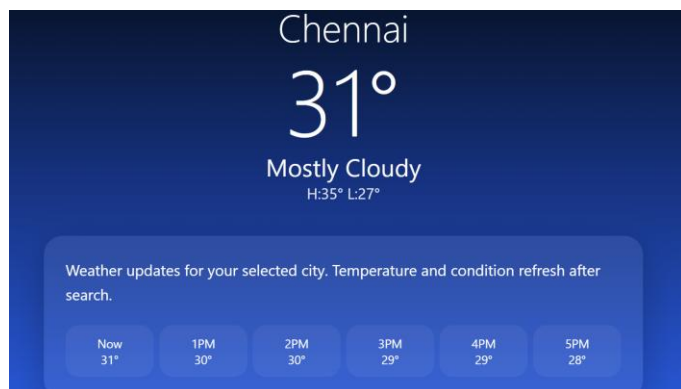
- Home Page
- Navigation Menu
- Login Interface
- Application Introduction



B. USER MODULE

Features:

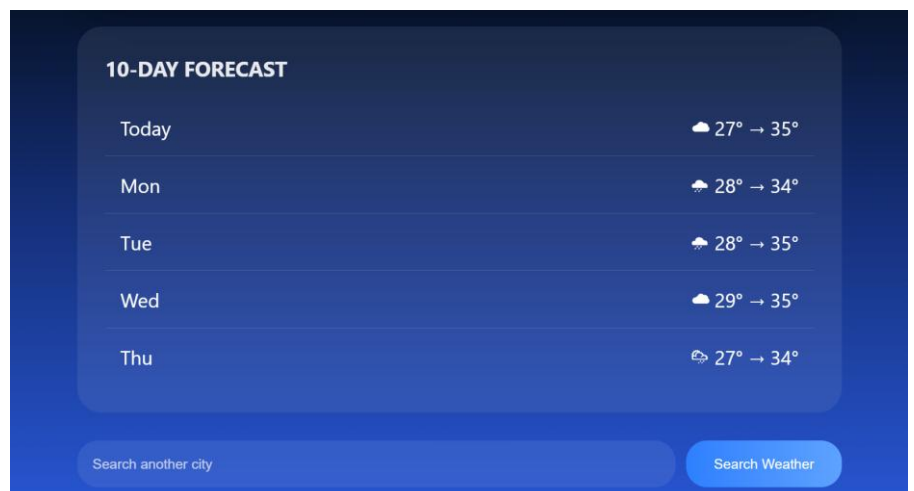
- User Registration
- User Login
- Personalized Dashboard
- Display Username
- Weather Search Interface



C. WEATHER SEARCH MODULE

Features:

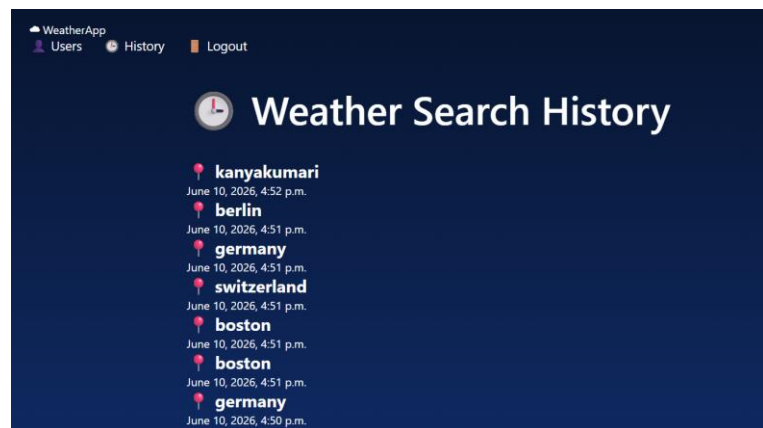
- Search Weather by City
- Current Temperature
- Maximum Temperature
- Minimum Temperature
- Humidity
- Wind Speed
- Weather Description



D. HISTORY MODULE

Features:

- View Previous Searches
- Store Search Date and Time
- Search History Display



E. AUTHENTICATION MODULE

Features:

- Login
- Logout
- Forgot Password
- Session Management

REQUIREMENTS

Hardware Requirements

Processor:

Intel Core i3 or Above

RAM:

4 GB Minimum

Storage:

20 GB Available Space

Software Requirements

Operating System:

Windows 10 / Windows 11

Programming Language:

Python 3.x

Framework:

Django

Frontend:

HTML

CSS

JavaScript

Database:

SQLite3

API:

OpenWeather API

IDE:

Visual Studio Code

Browser:

Google Chrome

DATABASE DESIGN

User Table

- Username
- Email
- Password

Weather Search Table

- User
- City Name
- Temperature
- Weather Condition
- Search Date

History Table

- User
- City
- Search Time

SYSTEM WORKFLOW

Step 1:

User opens application.

Step 2:

User creates account.

Step 3:

User logs into system.

Step 4:

Dashboard displays current weather.

Step 5:

User enters city name.

Step 6:

Application sends request to OpenWeather API.

Step 7:

API returns weather information.

Step 8:

Weather details displayed.

Step 9:

Search stored in history.

Step 10:

User logs out.

FEATURES IMPLEMENTED

Authentication

- ✓ Login
- ✓ Logout
- ✓ Create Account
- ✓ Forgot Password

Weather Features

- ✓ Search City Weather
- ✓ Current Temperature
- ✓ Weather Conditions
- ✓ Dynamic Weather Data

User Features

- ✓ Dashboard
- ✓ Responsive Interface
- ✓ Personalized Welcome

History Features

- ✓ Search History
- ✓ Date & Time Storage

PROJECT STRUCTURE

weatherforecast/

├── weatherforecast/

| ├── settings.py

| ├── urls.py

| ├── wsgi.py

├── weatherapp/

| ├── templates/

| ├── static/

| ├── views.py

| ├── models.py

├── users/

├── history/

├── db.sqlite3

└── manage.py

TECHNOLOGIES USED

Python

Django

HTML

CSS

JavaScript

SQLite3

OpenWeather API

VS Code

Git

ADVANTAGES

- Fast weather retrieval
- User-friendly UI
- Easy navigation
- Secure authentication
- Real-time weather updates
- Responsive design
- Search history tracking
- Lightweight architecture

FUTURE ENHANCEMENTS

- Weekly Weather Forecast
- GPS-based Location Detection
- Dark/Light Theme
- Weather Notifications
- Weather Maps
- Air Quality Index
- Mobile Application
- Multi-language Support
- Weather Analytics Dashboard

CONCLUSION

The Weather Forecast Web Application developed using Django successfully provides real-time weather information through a modern and responsive interface. The project demonstrates practical implementation of Django web development, API integration, authentication, and database handling.

The application allows users to search weather conditions efficiently while maintaining history and secure user management. Future improvements can transform this application into a complete commercial weather platform.